

Diagnostics Layer (RETAIN)

Failure Localization

FAMAS, TrajAudit,
DoVer, AgenTracer

Observability

Agent Traces, OTEL

Safety

G-Safeguard, ShieldAgent,
LlamaFirewall

Requires full per-step traces, snapshots,
cross-agent chains

SHARED RUNTIME STATE (RETENTION BUDGET)

Cross-layer conflict 1: Eviction

ScaleSim swaps out the state diagnosis needs

Cross-layer conflict 2: Snapshots

Replay needs environment state; serving discards it

Intra-layer analogue: SSF

TrajAudit compresses 86-92% (same tradeoff shape)

Boundary condition: Replay cost

FAMAS $k=20$ affordable only offline

Empirical evidence (TraceElephant)

Output-only capture costs 76% relative step-level attribution
accuracy

Infrastructure Layer (DISCARD)

Memory Mgmt

ScaleSim:
distance-aware eviction

Request Scheduling

TOPAS, TokenCake:
prefix-aware placement

Resource Allocation

Needs eviction, TTL expiry,
compression for fast service

Bounded by serving SLO; overhead is
waste